

Algorithmic Pseudocode Sample 50

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

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1: Input:
    • Algebraic system of difference equations named  $\Sigma'$ 
    • Time measured data allowing prolongation of the system.
    • For  $\bar{\mu} = \mu_1, \dots, \mu_n$  the finite set of parameters, the data  $R_{\mu_i}$  of permissible intervals for each parameter value.

2: Output: Parameter values of  $\Sigma'$ .
3: procedure DETECT SOLVABILITY
4:   Define  $\Sigma_{prolong}$  as an indefinite time prolongation of  $\Sigma'$ .
5:   Redefine  $\Sigma' := []$ , an ‘empty system’ of no equations.
6:   Denote  $e_1, \dots$  the equations of  $\Sigma_{prolong}$ .
7:   Define  $J(t) = \left[ \frac{\partial e_i}{\partial \mu_j} \right]$  where  $e_i \in \Sigma', i = 1, \dots, r$ , or 0 if  $\Sigma'$  is empty.
8:    $i := 1$ 
9:   for  $rank(J(t)) := s < n$  i.e. is not of full rank, do
10:    procedure MOVE EQUATION
11:       $\Sigma' := \Sigma' \cup \{e_i\}$ 
12:       $\Sigma_{prolong} := \Sigma_{prolong} \setminus \{e_i\}$ 
13:      Compute  $rank(J(t))$  (note  $\Sigma'$  has updated)
14:      if  $rank(J(t)) < s + 1$  then  $\Sigma' := \Sigma' \setminus \{e_i\}$ 
15:      end if
16:       $i = i + 1$ 
17:      if  $rank(J(t)) < n$  then repeat procedure Move Equation.
18:      end if
19:    end procedure
20:  end for
21:  procedure BLACKBOX SOLVER AND FILTER Note Detect Solvability runs until  $J(t)$ 
    has full rank and outputs a polynomial system  $\Sigma'$  that has solutions. Run any
    algebraic solver.
22:    Filter solutions by intersecting solution set with  $R_{\mu_i}$ .
23:  end procedure
24: end procedure
```
